

TELCO CUSTOMER CHURN

Update 2

Predictive Modeling · Intervention Strategy · Business Value

Ethan Kwon · Jihun Jeong · Henry Kim · Kerri Wang

Business Value Through Integrative Analytics | CMU Tepper | 2026

Update 1 Recap & Hypothesis Verification

Update 1 Clusters

Early High-Risk

Churn: 67.3% | Rev@Risk: \$430K/wk

High-Value Bundled

Churn: 15.9% | Rev@Risk: \$103K/wk

Loyal Advocate

Churn: 1.6% | Rev@Risk: \$7K/wk

Stable Low-Usage

Churn: 5.8% | Rev@Risk: \$39K/wk

Hypotheses → Confirmed by Model

H1

Month-to-month contract → higher churn

✓ 45.8% vs 2.5%

H2

Fiber Optic → higher churn

✓ 40.7% vs 18.6%

H3

New customers (<12m) → higher churn

✓ 47.4% vs 11.9%

H4

Higher bundle count → lower churn

✓ Coef: -0.12

H5

Higher engagement index → lower churn

✓ Coef: -0.64

Signal Leakage Audit & Feature Taxonomy

✅ Observable Before Churn — Safe Predictors

Demographics:

gender, age, married, number_of_dependents

Account:

tenure_in_months, contract, offer, payment_method

Services:

phone_service, internet_type, online_security, streaming_*,
online_backup, device_protection, premium_tech_support

Financial:

monthly_charge, total_charges, cltv

Survey:

satisfaction_score (pre-churn survey)

Engineered:

service_bundle_count, tenure_segment, engagement_index, cluster

❌ Post-Hoc — EXCLUDED

churn_value

TARGET — outcome variable

churn_label

String version of target

customer_status

Encodes churn outcome

churn_score

Company model score
→ data leakage if used

churn_category

Only known after churn

churn_reason

Only known after churn

Predictive Models — Training & Evaluation

Logistic
Regression

AUC

0.9908

Decision
Tree

AUC

0.9828

Random
Forest

AUC

0.9883

XGBoost

AUC

0.9919

Class Imbalance (26.5% positive): Addressed via class_weight='balanced' across all models

Top-10% Lift (churners captured vs random)

Model	Top-10% Captures	Lift
Logistic Regression	37.4% of all churners	3.74×
Decision Tree	36.6% of all churners	3.66×
Random Forest	37.4% of all churners	3.74×
XGBoost ★	37.4% of all churners	3.74×

Intervention Value Framework

$$v_{ij} = \hat{p}_i \cdot \eta_j \cdot \text{CLTV}_i - c_j$$

$\hat{p}_i = \text{Pr}(\text{churn})$ | $\eta_j = \text{retention success rate}$ |
 $c_j = \text{intervention cost}$

Tier 1

Email / SMS

Cost (c_j)

\$20

Retention Rate (η_j)

10%

Positive-value customers

542 customers (38.5%)

Mean v_{ij}

\$97.8

Tier 2

Phone Outreach

Cost (c_j)

\$100

Retention Rate (η_j)

25%

Positive-value customers

497 customers (35.3%)

Mean v_{ij}

\$194.6

Tier 3

Premium Package

Cost (c_j)

\$300

Retention Rate (η_j)

45%

Positive-value customers

467 customers (33.1%)

Mean v_{ij}

\$230.3

Tier assumptions stated — no external data provided. Best tier per customer assigned based on highest positive v_{ij} .

Heuristic Strategies — Which Approach Makes Most Money?

Budget: top 30% of customers (422 out of 1,409 test set)

S1: Random



\$180K

116 churners

S2: Top Pr(churn)



\$580K

358 churners

S3: Revenue at Risk



\$582K

354 churners

S4: Multi-tier Greedy



\$580K

358 churners

S5: Top v_{ij} ★



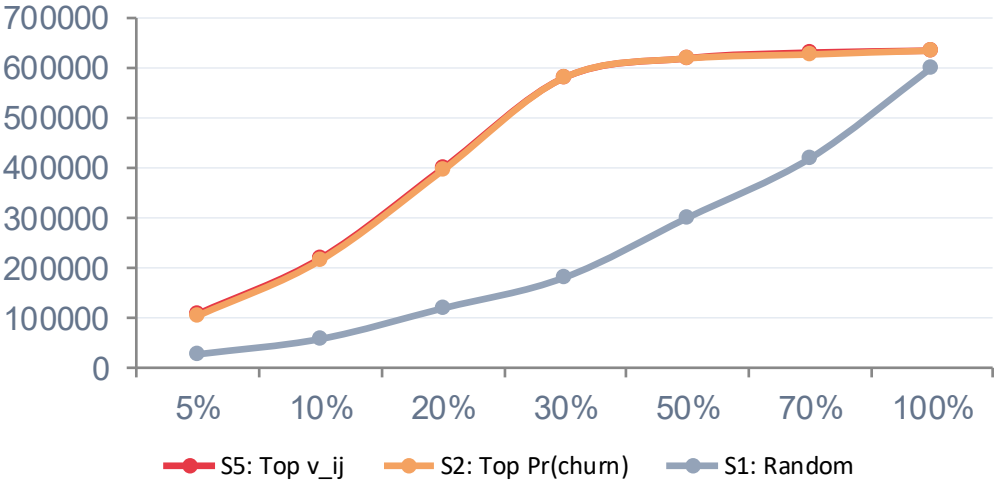
\$582K

367 churners

All model-based strategies deliver 3.2× more value than random — confirming that prediction quality directly drives business impact.

Value Scaling & Cluster-Level Strategy Performance

Value vs Budget (% Customers Targeted)



Cluster Uncertainty — Std of $\text{Pr}(\text{churn})$

Cluster	Mean $\hat{p}$	Std($\hat{p}$)
Early High-Risk	0.687	± 0.422
High-Value Bundled	0.195	± 0.352
Stable Low-Usage	0.058	± 0.193
Loyal Advocate	0.022	± 0.120

Large std in Cluster 0 = high prediction uncertainty
→ warrants broader screening net

S5 (Best Strategy) — Customers Selected by Cluster

Early High-Risk n=306 281 churners \$434K	High-Value Bundled n=79 55 churners \$113K	Stable Low-Usage n=36 17 churners \$33K	Loyal Advocate n=1 1 churners \$2K
---	--	---	--

Temporal Validation & Model Robustness

Validation Approach

Train: Established customers (tenure > 36 months) — 3,001 customers, churn rate 11.9% **Test:** New/Developing customers (tenure ≤ 36 months) — 4,042 customers, churn rate 37.4%

Random Split AUC
(stratified 80/20)

0.9883

Standard cross-validation

Temporal Split AUC
(established → new)

0.9785

Out-of-time validation

Performance Gap

0.0098

*Model generalizes well
across tenure cohorts*

Top Features (Random Forest & XGBoost) — Hypothesis Alignment

Rank	Random Forest	XGBoost	Hypothesis Link
#1	satisfaction_score (0.420)	satisfaction_score (0.336)	Validates PCA PC2
#2	cluster (0.162)	cluster (0.334)	Validates Update 1 segmentation
#3	contract_enc (0.094)	online_security (0.076)	H1 confirmed
#4	engagement_index (0.044)	contract_enc (0.030)	H5 confirmed

Managerial Summary

& Strategic Recommendations

1

Model quality is high — XGBoost AUC 0.99

All models generalize well across cohorts (temporal gap: 0.010). Top-decile lift = 3.74× over random.

2

Prediction alone is not enough — use Intervention Value

Strategy S5 (top v_ij) and S3 (revenue at risk) both deliver \$582K vs \$180K for random — a 3.2× improvement.

3

Cluster 0 (Early High-Risk) deserves immediate action

Captures 73% of all value in Strategy S5. Std of Pr(churn) = 0.42 → broad Tier 1 outreach is warranted.

4

Protect Cluster 1 (High-Value Bundled) with premium tiers

Highest CLTV (\$5,025). Even moderate churn = disproportionate loss. Tier 3 (\$300 intervention) is economically justified.

Tier Recommendation: Cluster 0 → Tier 1 (broad) | Cluster 1 at-risk → Tier 3 (targeted) | Others → monitor